

Assignment - 3

Combinational Assignment

1) Implement the following functions using
 i) 2:1 MUX and NOT gates ii) 4:1 MUX

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

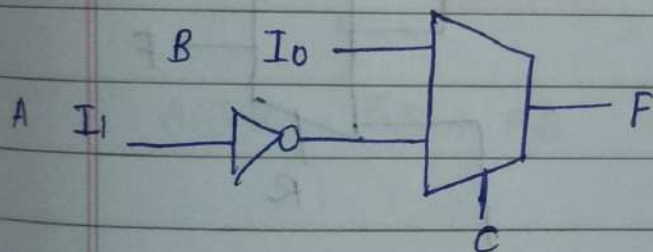
b) $F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

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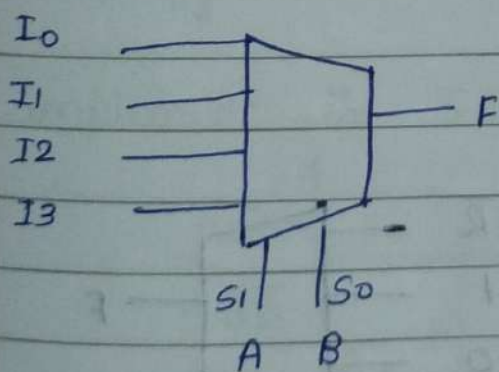
i) 2:1 MUX and NOT gates

When $C = 0$ $F = B$

$C = 1$ $F = \bar{A}$



ii) 4:1 MUX



A	B	F
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$

2) Implement the functionality of a full adder using 2:1 Muxes only.

$$b) F(P, Q, R) = \bar{P}\bar{R} + QR + \bar{P}Q$$

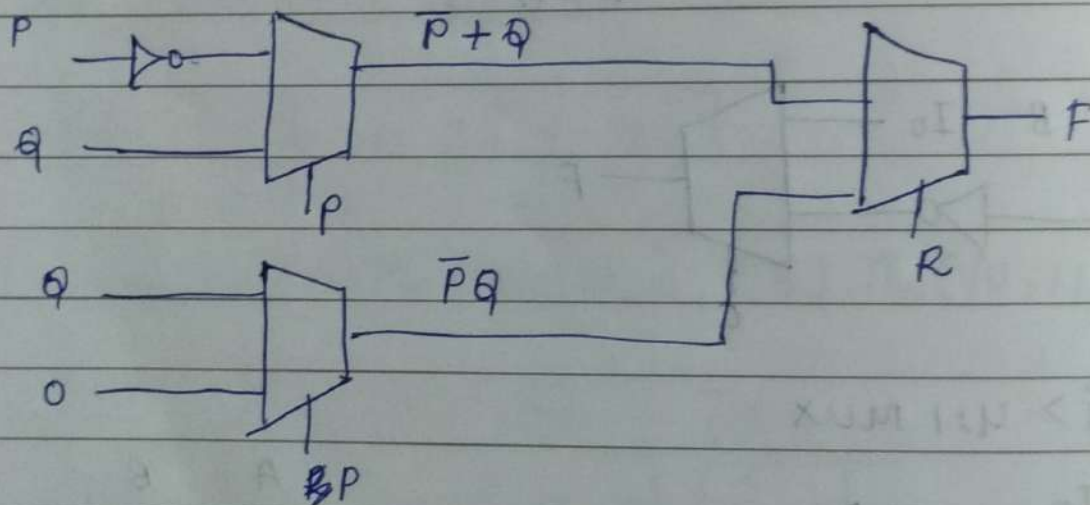
iz 2:1 Mux and NOT gates

$$\text{when } R=0 \quad F = \bar{P} + Q + \bar{P}Q = \bar{P} + Q$$

$$R=1 \quad F = \bar{P}Q$$

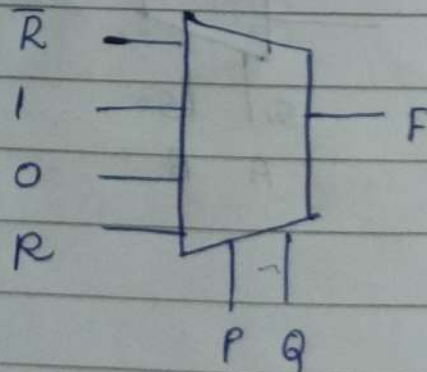
$$I_0 = \bar{P} + Q$$

$$I_1 = \bar{P}Q$$

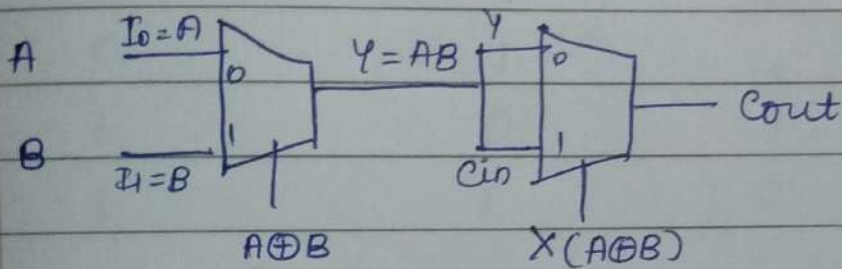
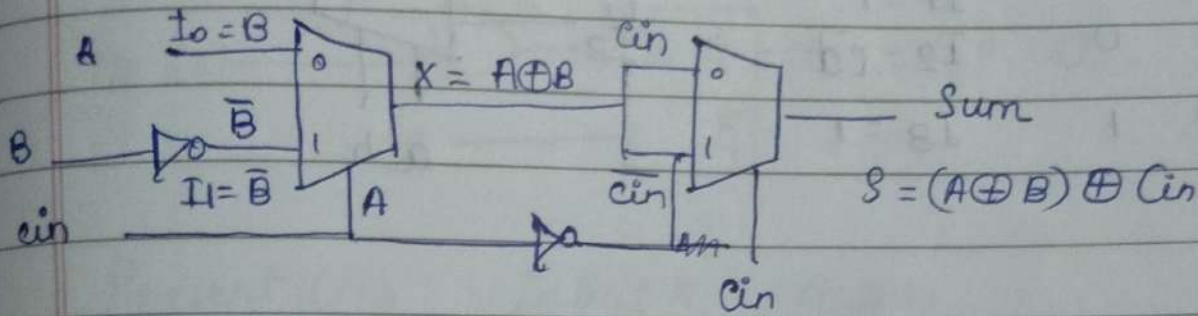


ii) Using 4:1 Mux

P	Q	F
0	0	$\bar{R}$
0	1	1
1	0	0
1	1	R



→ Full adder :- Sum $A \oplus B \oplus C$
Carry $AB \oplus Cin (A \oplus B)$



$$A \oplus B = \bar{A}B + A\bar{B} \quad \text{when } A=1, O/P = \bar{B}$$

$$A=0 \quad O/P = B$$

3) Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) +$
 $D(12, 13, 14, 15)$ and

implement the SOP expression using 4:1 Mux

	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0	1	3	2
$\bar{a}b$	4	5	7	6
ab	X ₁₂	X ₁₃	X ₁₅	X ₁₄
$a\bar{b}$	8	9	11	10

$$f = b\bar{c} + b\bar{d} + c\bar{d}$$

$$= b(\bar{c} + \bar{d}) + c\bar{d}$$

a	b	Mux I/P
0	0	$I_0 = c\bar{d}$
0	1	$I_1 = 1$
1	0	$I_2 = c\bar{d}$
1	1	$I_3 = 1$

